MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPE 3FP7A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	Vdc	v	Vdc	Vdc	Vdc	Meg	Meg	Vdc	Ratio	ft
Maximum:	6.9	0	550	1,100	2,200	4,400	1.5	1.0	-125	2.3	10,000
Minimum:	5.7	-125	---	---	1,500	3,000	---	---	---	---	---
TEST CONDITIONS:	6.3	Adj	---	Focus	2,000	4,000	---	---	---	---	---

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection						
5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	2.25	Inch
5213	Cathode illumination	-		---	---	---	---
5246	Focusing voltage	-		Eb1	400	690	Vdc
5248	Deflection factor (1D2)	-	Eb3 = Eb2 = 2,000 Vdc	DF	170	230	Vdc/ in.
5248	Deflection factor (3D4)	-	Eb3 = Eb2 = 2,000 Vdc	DF	126	170	Vdc/ in.
5248	Deflection factor uniformity	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Control grid to all	Cg1	---	10.5	pF
			Cathode to all	Ck	---	10.5	pF
			D1 to D2	C1D2	---	4.0	pF
			D3 to D4	C3D4	---	4.0	pF
			D1 to all	CD1	---	15.0	pF
			D3 to all	CD3	---	12.0	pF
			D1 to all except D2	CD1	---	13.0	pF
			D2 to all except D1	CD2	---	13.0	pF
			D3 to all except D4	CD3	---	12.0	pF
			D4 to all except D3	CD4	---	12.0	pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
(A)1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	1	Ib3 = 150 µAdc	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5225	Modulation	-	Ib3 = 150 µAdc	ΔEc	---	50	Vdc
5251	Spot position (electrostatic deflection)	-		---	---	12	mm
5251	Spot displacement (leakage)	-		Displ	---	7	mm
5241	Grid cutoff voltage	-		Eco	-50	-90	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 -Continued						
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-50	10	µAdc
5201	Electrode current (cathode)	-	Ib3 = 150 µAdc	Ik	---	1,500	µAdc
5101	Base alinement (electrostatic types)	-	+1D2; pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Angle between traces	-		---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 2,200 Vdc; Eb3 = 4,400 Vdc	---	---	---	---
5221	Screens	-		---	---	---	---
5226	Line width "A" (electrostatic deflection)	2	Ib3 = 150 µAdc	Width	---	0.75	mm
5226	Line width "B" (electrostatic deflection)	3	Ib3 = 150 µAdc	Width	---	0.90	mm
5246	Focusing voltage at cutoff	-		Eb1	460	690	Vdc
5248	Deflection factor	-	1D2	DF	212	288	Vdc/in.
5248	Deflection factor	-	3D4	DF	153	207	Vdc/in.
5251	Heater-cathode leakage current	-		---	---	---	---
(A)1101	Secureness of base, cap, or insert	-		---	---	---	---
(A)1111	Base pin solder depth	-		---	---	---	---
(A)1105	Permanence of marking	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb3 = 4,400 Vdc; Eb2 = 2,200 Vdc; Ib3 = 30 µAdc; t = 500 hours	---	---	---	---
---	Life-test end points:						
5223	Modulation	-	Ib3 = 112 µAdc	ΔEc	---	50	Vdc
5226	Line width "A"	-	Ib3 = 112 µAdc	Width	---	0.8	mm
5226	Line width "B"	-	Ib3 = 112 µAdc	Width	---	0.95	mm

NOTES:

1. This test to be performed at the conclusion of the holding period.

2. The deflecting plates shall be returned to anode number 2 through a minimum of 2.5 megohm resistors. Ib3 the beam current shall be set at 150 µAdc. The high-frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus.
   (a) The tubes shall be observed for deflection defocusing, astigmatism or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of trace (usually around edges) bow-tieing (irregular widths of any single line when observed at different points) bowing of trace other than that normally caused by curvature of bulb.
   (b) This test for focus is to be made in addition to the line width measurements.

3. The same conditions shall be setup as described in note 2, except that the connection of deflecting elements to the low- and high-frequency scanning supplies shall be inter-changed and the amplitudes adjusted to 90 percent of focus from maximum tube diameter in both directions without any adjustment of focus from note 2 conditions. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 2.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Navy - EC
Air Force - 17, 80

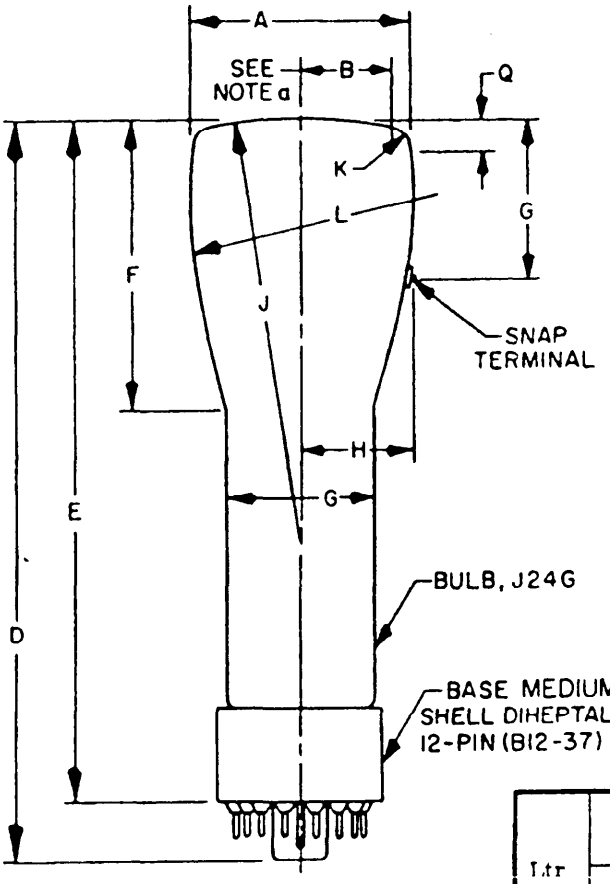
User activities:
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Navy - AS, OS, MC, CG, SH
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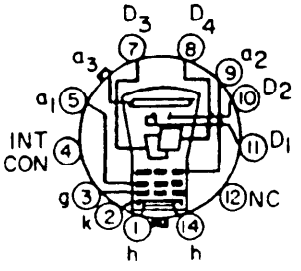
Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.937 (74.60) Dia.	3.063 (77.80) Dia.
B	1.375 (34.93)	
C	1.937 (49.20) Dia.	2.063 (52.40) Dia
D	9.750 (247.65)	10.250 (260.35)
G	1.500 (38.10)	2.000 (50.80)
Reference dimensions (see note b)		
E	9.250 (234.95)	
F	3.875 (98.43)	
H	1.625 (41.28)	
J	8.000 (203.20) Rad.	
K	.250 (6.35) Rad.	
L	12.438 (315.93) Rad.	
Q	.350 (8.89)	

NOTES:
a. Minimum useful screen radius.
b. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube 3FP7A.

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